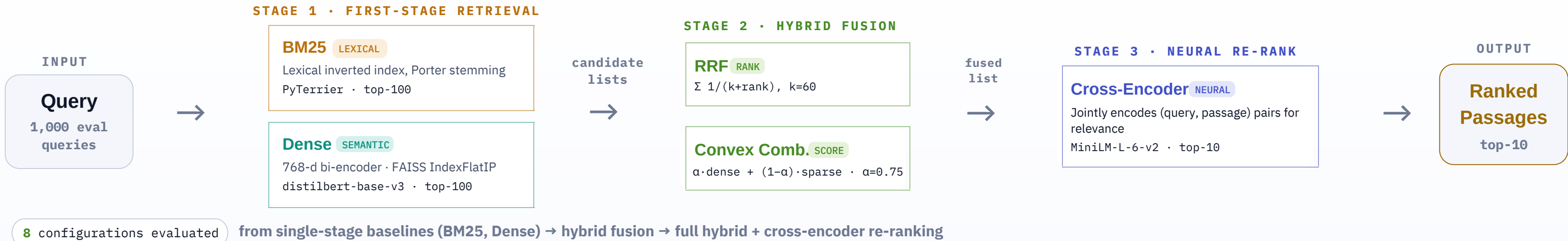


Hybrid Passage Retrieval

Lexical × Semantic × Neural

A four-stage retrieval pipeline that fuses **BM25 lexical search**, a **dense bi-encoder**, **hybrid score fusion**, and a **cross-encoder re-ranker** — benchmarked across **8 end-to-end configurations** on MS MARCO Passage and TREC Deep Learning 2019.

01 METHODOLOGY – THE RETRIEVAL PIPELINE



02 MOTIVATION

Lexical search matches exact terms — fast and precise, but blind to meaning.
Semantic search captures intent, but can drift from the literal query.

Neither approach wins alone. We propose a **multi-stage pipeline** that merges the high recall of efficient first-stage retrieval with the exact precision of a neural re-ranker to achieve state-of-the-art performance.

04 KEY FINDINGS

- Dense dominates BM25** on both benchmarks, confirming semantic matching's edge.
- Optimized hybrid pipelines improve the system** by intelligently merging exact keyword matches with semantic context.
- Convex Combination beats RRF** at first-stage fusion (0.616 vs 0.580 MRR@10) — score-level > rank fusion.
- Re-ranking is the biggest lever** — a cross-encoder lifts every retriever; even BM25+CE beats standalone Dense.

03 RESULTS – EVALUATION

MS MARCO Passage

dev/small · binary · 1,000 query

SYSTEM	MRR@10	NDCG@10
BM25	0.4517	0.4925
Dense	0.5913	0.6335
Hybrid RRF	0.5801	0.6241
Hybrid CC	0.6164	0.6584
BM25 + Re-ranking	0.6849	0.7135
Dense + Re-ranking	0.7126	0.7475
Hybrid RRF + Re-ranking	0.7197	0.7563
Hybrid CC + Re-ranking ★	0.7201	0.7566

lexical dense hybrid + cross-encoder

TREC-DL 2019

8 configurations · trec_eval-compatible (ir_measures)
graded 0-3 · 43 query

SYSTEM	MRR≥2@10	NDCG@10
BM25	0.7359	0.6235
Dense	0.9147	0.7435
Hybrid RRF	0.8934	0.7354
Hybrid CC	0.9018	0.7576
BM25 + Re-ranking	0.9089	0.7798
Dense + Re-ranking	0.9205	0.7980
Hybrid RRF + Re-ranking	0.9205	0.8108
Hybrid CC + Re-ranking ★	0.9205	0.8120

★ best system